



# Final Project — Expanded Data Engineering Capstone

Path A: Official AWS Academy Capstone, expanded

**Subject:** Big-Data

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**Rol 1:** Data Engineer

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**Rol 4:** Orchestration & Operations

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## | Context: Sea Around Us (SAU)



### The Challenge of Marine Data

We work with the global fisheries dataset spanning 1950–2018. This data is critical for understanding the human impact on marine ecosystems across international waters.

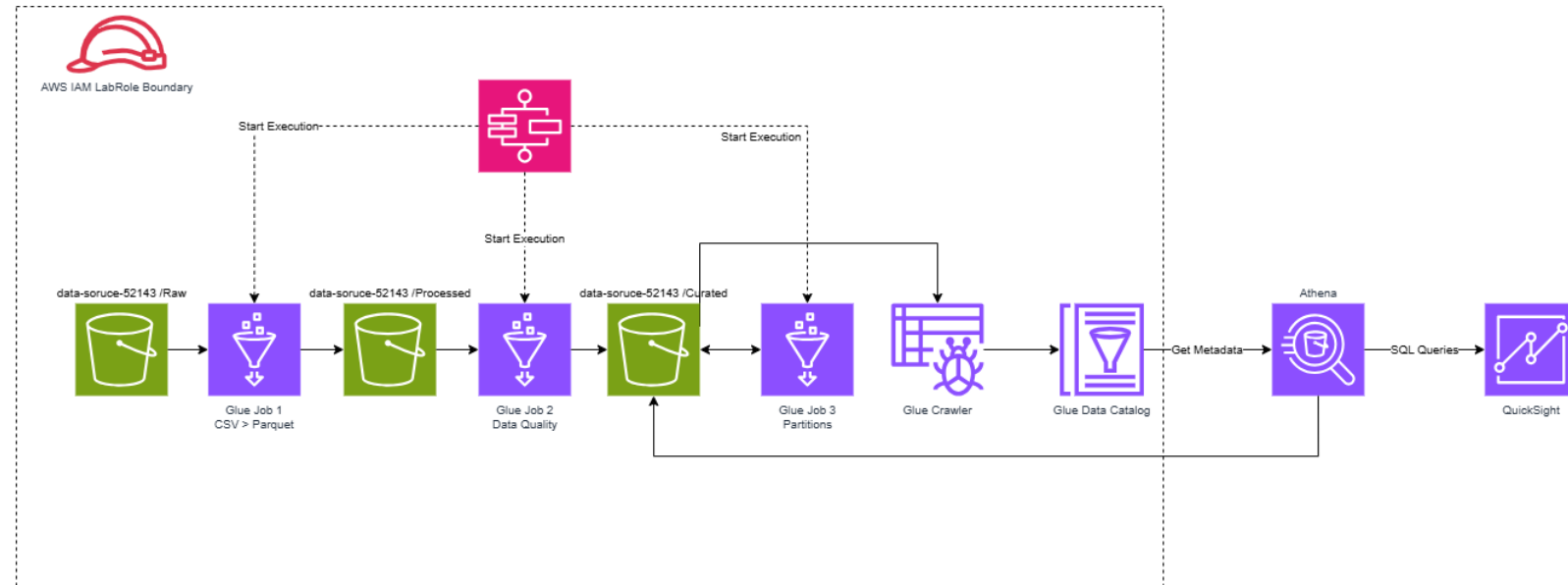
- **Sources:** Global catch, High Seas, EEZ Fiji, EEZ 848
- **Volume:** 500,000+ catch records across 68 years
- **Goal:** Transform fragmented CSV files from multiple ocean zones into a fully optimized, query-ready Data Lake on AWS

# Architecture

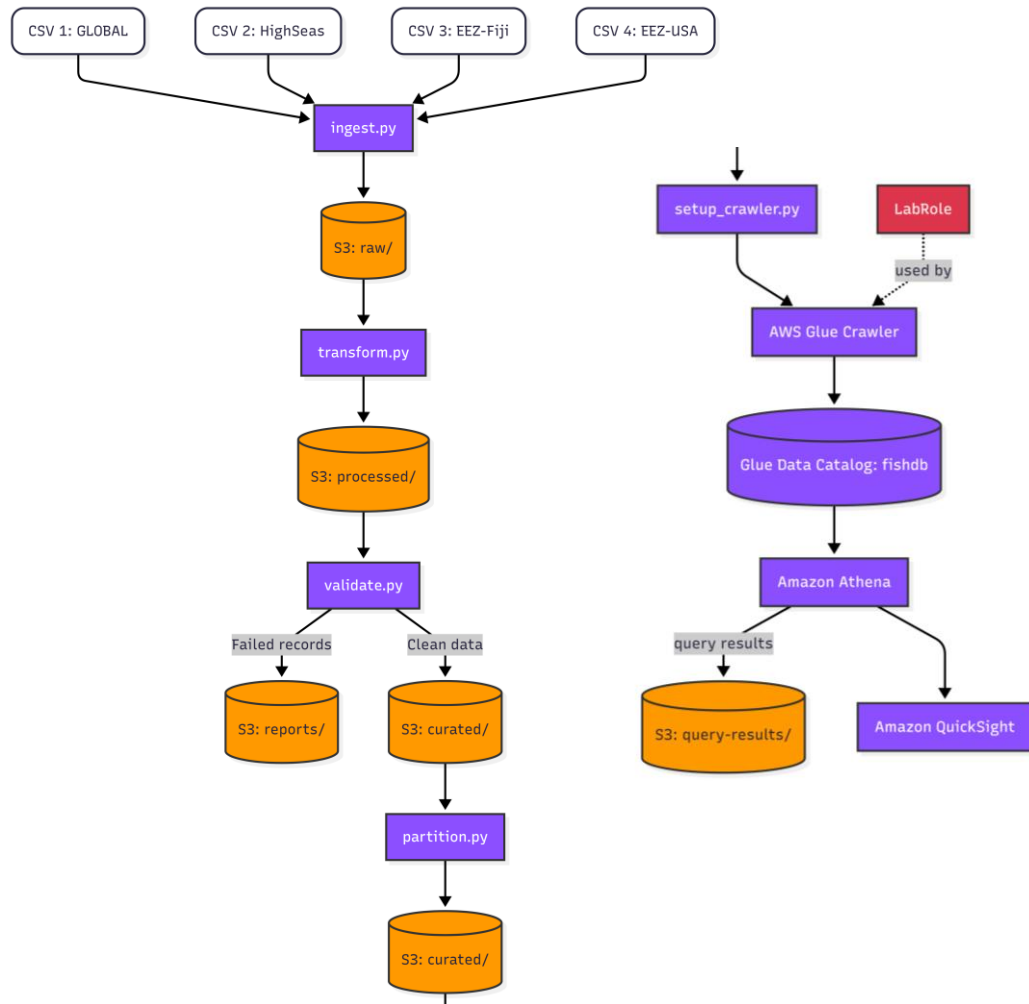
## Orchestration

Our solution is 100% Serverless. **AWS Step Functions** acts as the brain of the system, triggering each Glue Job in sequence and ensuring no stage runs before the previous one completes.

- **S3 Data Lake:** Three isolated zones — data-source-52143/Raw → /Processed → /Curated
- **Glue Job 1:** Ingests raw CSVs and converts them to Parquet format
- **Glue Job 2:** Applies data quality rules before data reaches the curated zone
- **Glue Job 3:** Applies partitioning to optimize downstream query performance
- **Glue Crawler:** Scans the processed bucket and registers schema metadata into the Glue Data Catalog
- **Athena:** Queries data directly on S3 via SQL — no data movement required
- **QuickSight:** Connects to Athena to deliver the final analytics dashboard



# End-to-end pipeline



## Zones

- **Raw:** original csv files
- **Processed:** cleaned parquet files with aligned schemas
- **Curated:** quality-validated and partitioned by year

```
s3://data-source-52143/  
├ raw/  
├ processed/  
└ curated/
```

## Scripts

### ingest.py

- Creates a bucket named data-source-52143 if it does not already exist.
- Uploads the four datasets to the raw/ zone in S3.
- No data is modified at this stage.

### transform.py

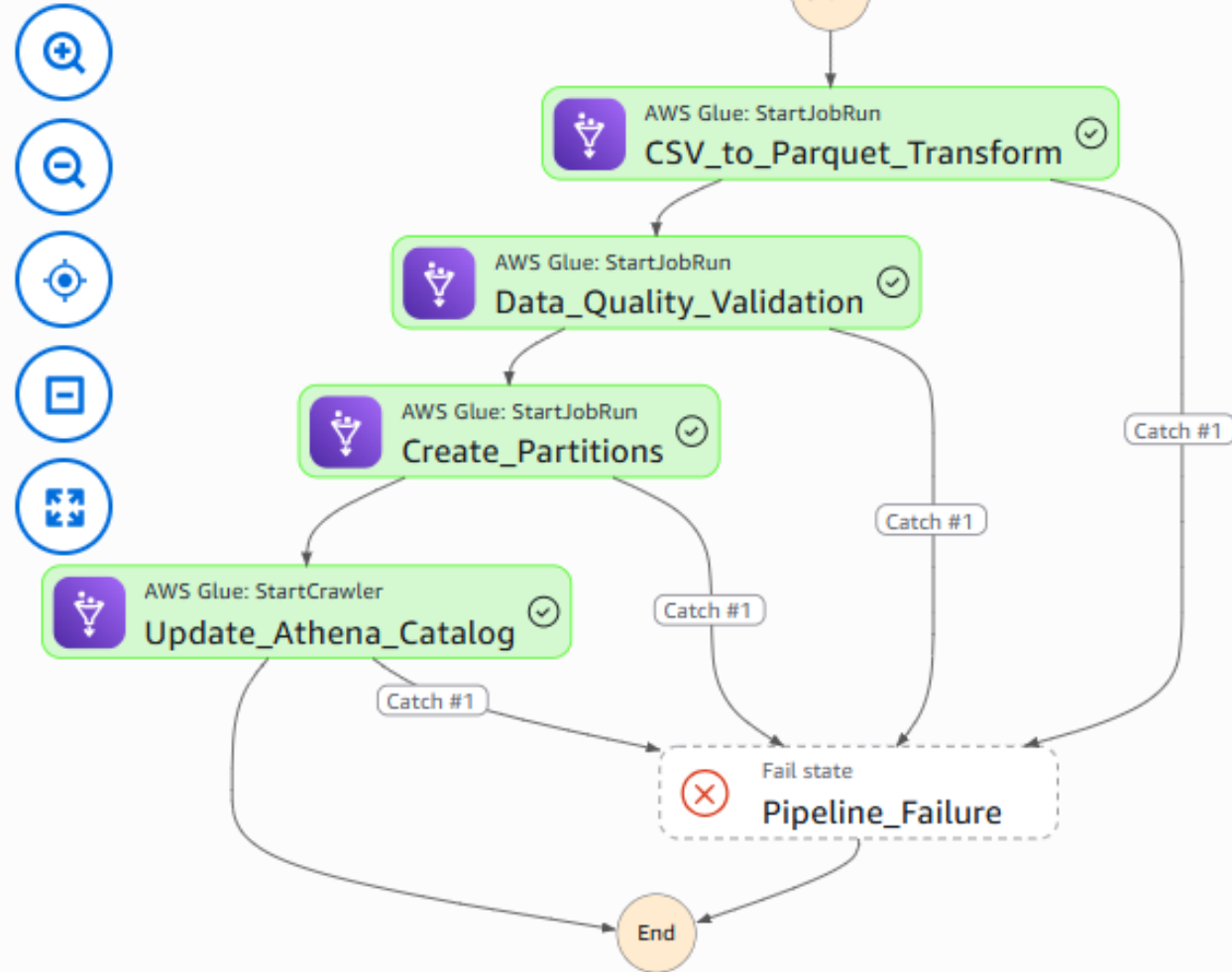
- Applies schema alignment where needed, converts to parquet format, and uploads the result to the processed/ zone.

### setup\_crawler.py

- Creates the fishdb Glue database and fishcrawler crawler if they do not already exist, then runs the crawler against the curated/ zone (after Role 2 validates the data and Role 3 partitions it) and waits for it to complete. Prints the list of tables created in the Glue Data Catalog.

## Graph view

Actions ▼



In progress



Failed



Caught error



Canceled



Succeeded

| ID    | Rule                                        | Files               | Severity |
|-------|---------------------------------------------|---------------------|----------|
| DQ-01 | Year range 1950–2018                        | All (EEZ-848 fails) | Critical |
| DQ-02 | Tonnes $\geq 0$                             | All                 | Critical |
| DQ-03 | Nulls in landed_value $< 20\%$              | EEZ-848 at limit    | Critical |
| DQ-04 | Column rename applied correctly             | EEZ-242, EEZ-848    | Critical |
| DQ-05 | Uncertainty score in valid range            | EEZ-242, EEZ-848    | Warning  |
| DQ-06 | Consistency between tonnes and landed_value | All                 | Warning  |
| DQ-07 | Referential integrity of countries          | HighSeas, EEZs      | Warning  |
| DQ-08 | No ghost tonnes ( $< 1e-6$ )                | EEZ-848 mainly      | Info     |

# Data Quality Report

Run: 2026-05-18 06:02:26 | Sea Around Us — Fisheries Dataset

29  
Rules run

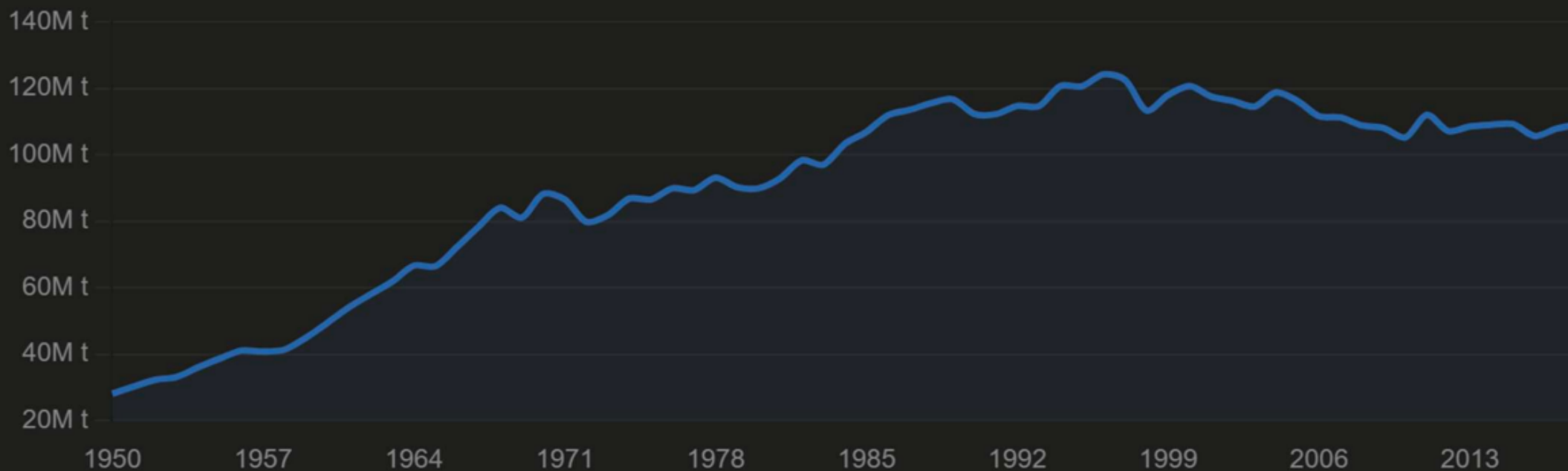
28  
Passed

1  
Failed

| ID    | Description                                          | File     | Status | Failed records | Total records | Failed % | Severity | Detail      |
|-------|------------------------------------------------------|----------|--------|----------------|---------------|----------|----------|-------------|
| DQ-01 | Year range 1950–2018                                 | GLOBAL   | PASSED | 0              | 561675        | 0.00%    | Critical |             |
| DQ-01 | Year range 1950–2018                                 | HighSeas | PASSED | 0              | 26720         | 0.00%    | Critical |             |
| DQ-01 | Year range 1950–2018                                 | EEZ_Fiji | PASSED | 0              | 27049         | 0.00%    | Critical |             |
| DQ-01 | Year range 1950–2018                                 | EEZ_848  | FAILED | 1746           | 83175         | 2.10%    | Critical | [2019]      |
| DQ-02 | Tonnes >= 0                                          | GLOBAL   | PASSED | 0              | 561675        | 0.00%    | Critical |             |
| DQ-02 | Tonnes >= 0                                          | HighSeas | PASSED | 0              | 26720         | 0.00%    | Critical |             |
| DQ-02 | Tonnes >= 0                                          | EEZ_Fiji | PASSED | 0              | 27049         | 0.00%    | Critical |             |
| DQ-02 | Tonnes >= 0                                          | EEZ_848  | PASSED | 0              | 83175         | 0.00%    | Critical |             |
| DQ-03 | Nulls in landed_value < 20%                          | GLOBAL   | PASSED | 0              | 561675        | 0.00%    | Critical | 0.0% nulls  |
| DQ-03 | Nulls in landed_value < 20%                          | HighSeas | PASSED | 0              | 26720         | 0.00%    | Critical | 0.0% nulls  |
| DQ-03 | Nulls in landed_value < 20%                          | EEZ_Fiji | PASSED | 0              | 27049         | 0.00%    | Critical | 0.0% nulls  |
| DQ-03 | Nulls in landed_value < 20%                          | EEZ_848  | PASSED | 15289          | 83175         | 18.38%   | Critical | 18.4% nulls |
| DQ-04 | Rename fish_name→common_name, country→fishing_entity | GLOBAL   | PASSED | 0              | 4             | 0.00%    | Critical |             |
| DQ-04 | Rename fish_name→common_name, country→fishing_entity | HighSeas | PASSED | 0              | 4             | 0.00%    | Critical |             |
| DQ-04 | Rename fish_name→common_name, country→fishing_entity | EEZ_Fiji | PASSED | 0              | 4             | 0.00%    | Critical |             |
| DQ-04 | Rename fish_name→common_name, country→fishing_entity | EEZ_848  | PASSED | 0              | 4             | 0.00%    | Critical |             |

## v1 · Tendencia de capturas globales por año

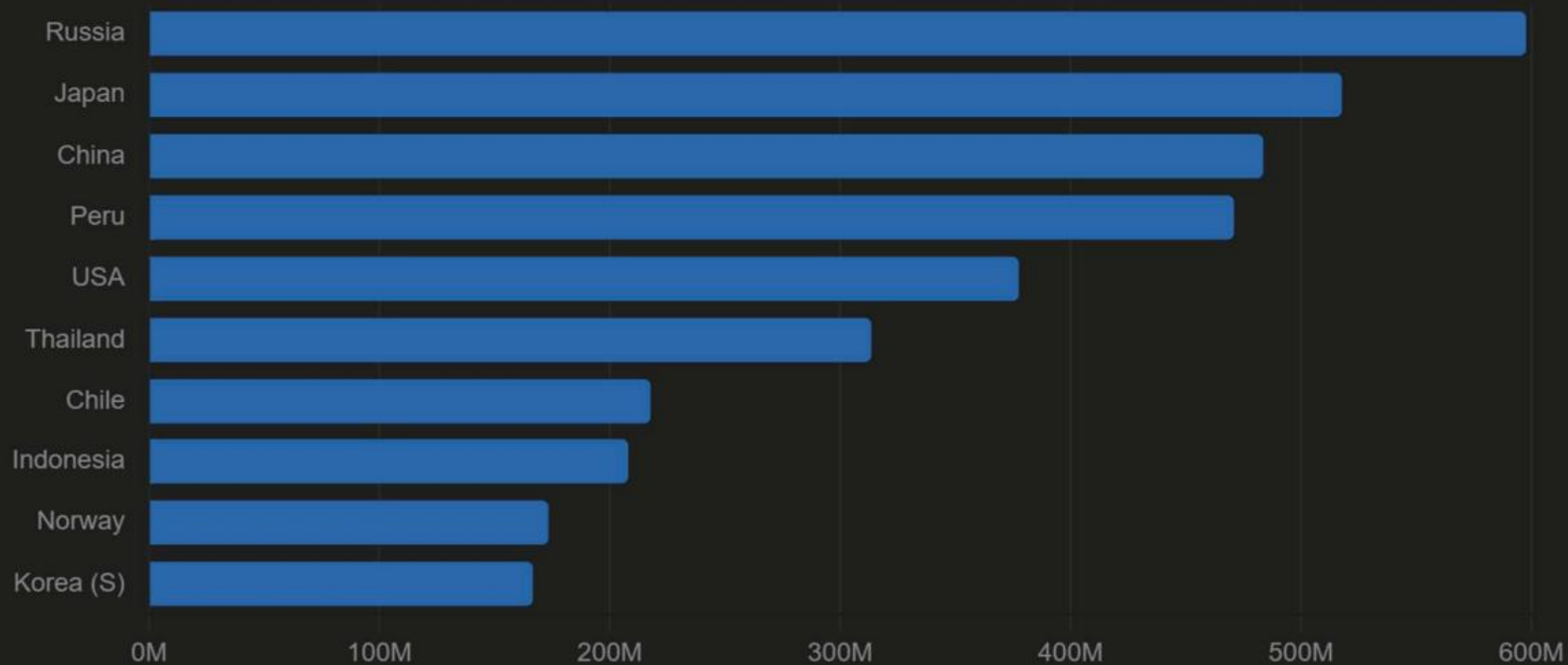
Millones de toneladas · 1950-2018 · fuente: sau\_global\_1





## v2 · Top 10 entidades pesqueras históricas

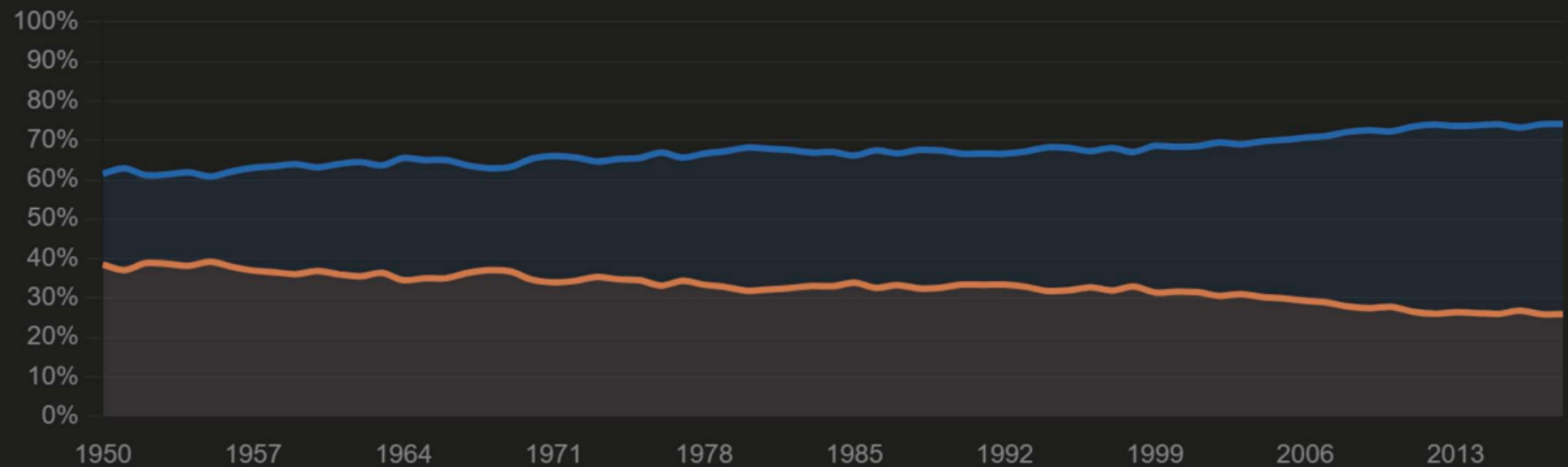
Millones de toneladas totales acumuladas · fuente: sau\_eez\_848



### v3 · Capturas reportadas vs no reportadas por año

Porcentaje del total anual · fuente: sau\_global\_1

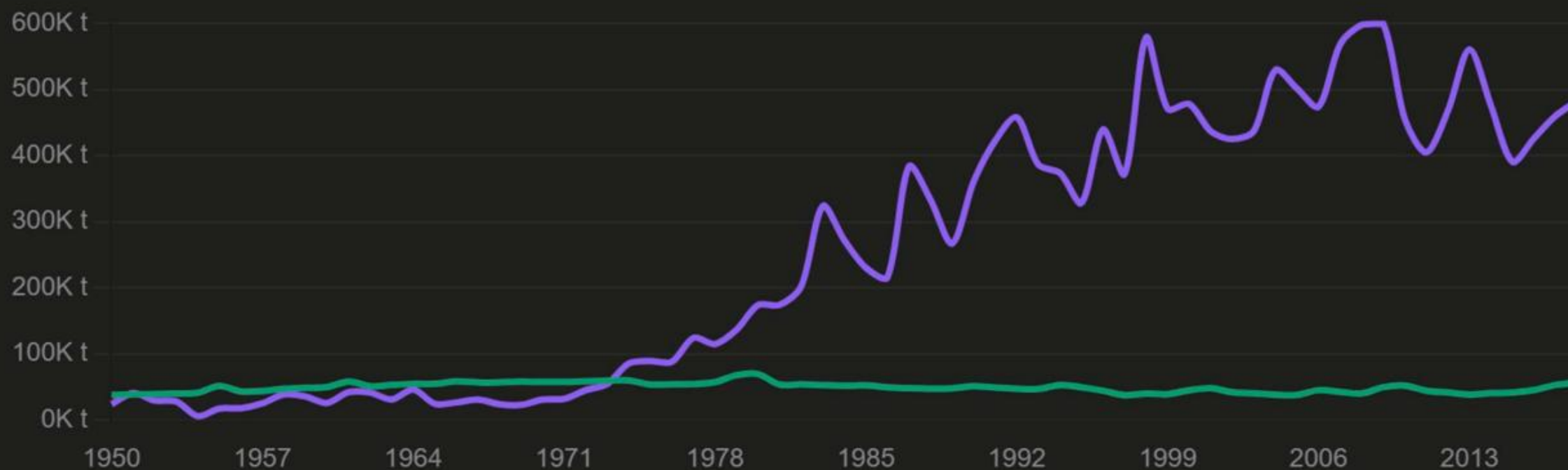
Reportado No reportado



## v4 · EEZ (242) vs High Seas por año

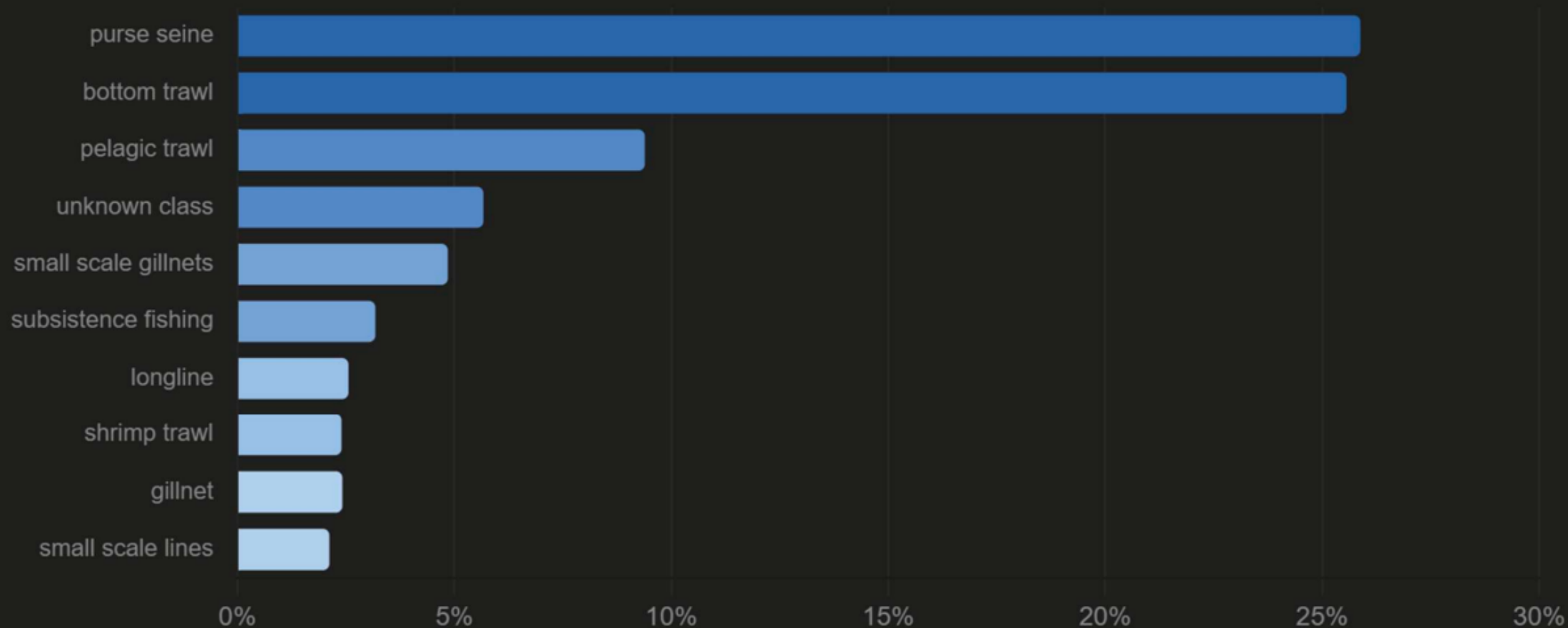
Miles de toneladas · fuente: sau\_eez\_242 + sau\_highseas

■ EEZ (242) ■ High Seas



## v5 · Top artes de pesca por % de captura total

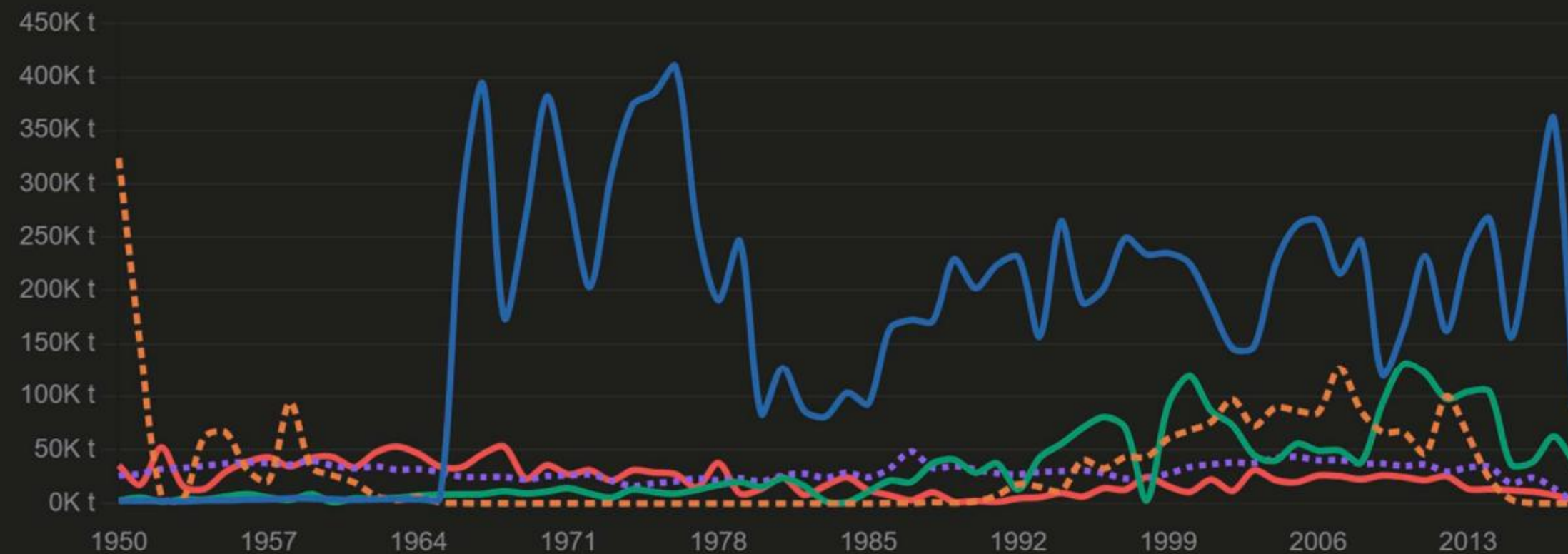
Solo artes con más de 1% del total · fuente: sau\_highseas



## v6 · Tendencia de top 5 especies por año

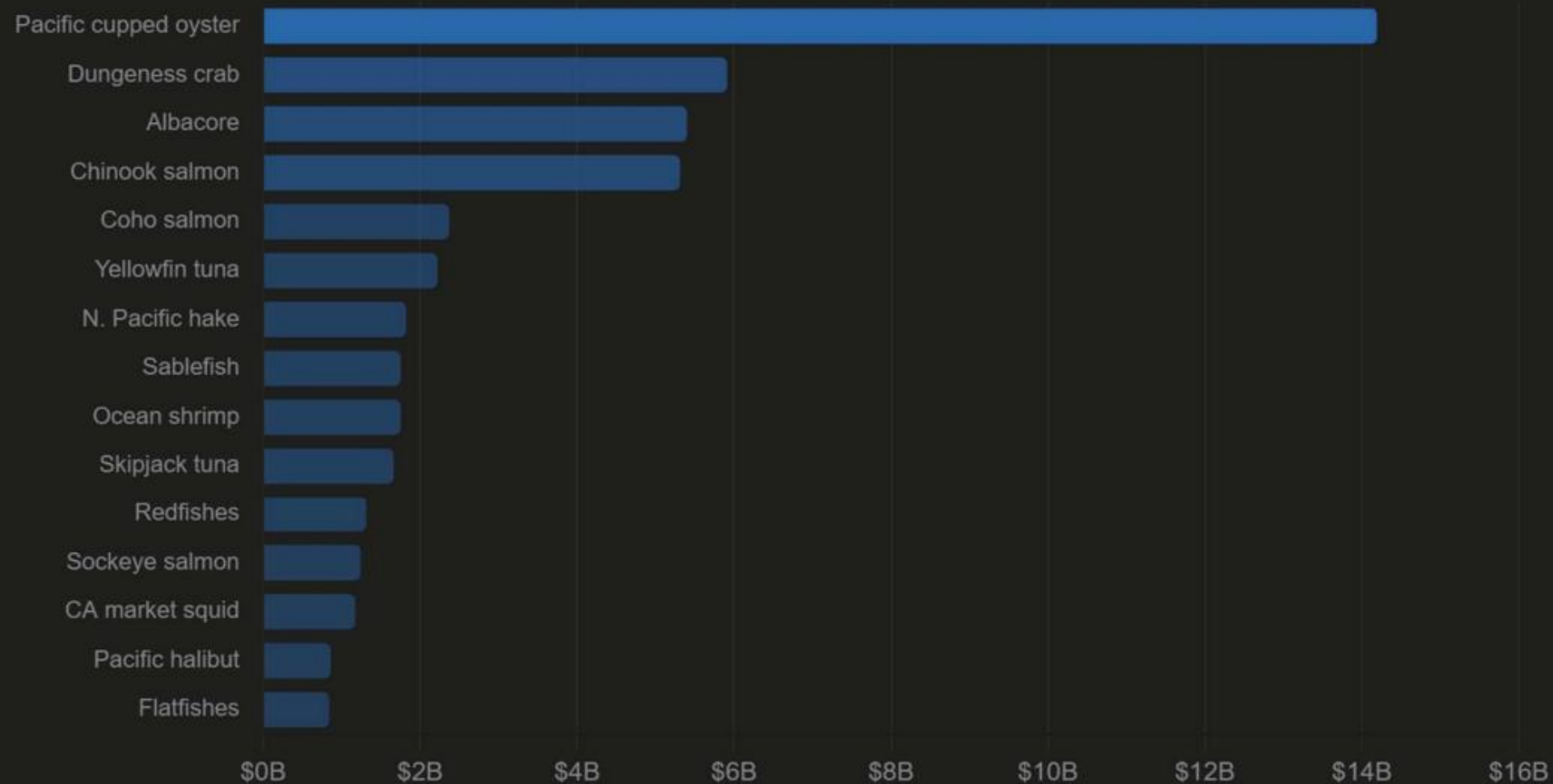
Miles de toneladas · fuente: sau\_eez\_848

■ N. Pacific hake ■ S. American pilchard ■ CA market squid ■ Pacific oyster ■ Albacore



## Valor económico total por especie

Miles de millones de USD acumulados · fuente: sau\_eez\_848



## Valor por tonelada — qué especie es más rentable por kg

USD por tonelada · fuente: sau\_eez\_848

